PW SKILLS

DDL, DML AND CONSTRAINTS

Q1. Create a database named CompanyDB. Create a table named Employees with the following columns:

Column	Data Type
Employee_ID	INT
Employee_Name	VARCHAR(50)
Salary	DECIMAL(10,2)

Q2. A startup is building an Employee Management System. Create a table named Employee with the following requirements:

- Employee ID must uniquely identify each employee.
- Employee Name cannot be empty.
- Email addresses should not be duplicated.
- Joining Date should automatically store today's date if not provided.

Q3. A company's HR department has announced a 10% salary hike for all employees.

The following Employee table already exists:

Column Name	Data Type
EmpID	INT
Name	VARCHAR(100)
Department	VARCHAR(50)
Salary	DECIMAL(10,2)
Experience	INT

Sample Data

EmpID	Name	Department	Salary	Experience
101	Alice	HR	50000	5
102	Bob	IT	60000	3
103	Charlie	Finance	55000	8
104	David	HR	58000	2
105	Edward	IT	62000	6

Task

1. Create the Employee table using suitable data types.

2. Insert the given records.

3. Write an SQL query to increase the salary of every employee by 10%.

Write the query to update the salary column

Q4. Hospital Management System

A hospital wants to maintain patient records.

Requirements:

- Patient ID should be unique.
- Age must be between 0 and 120.
- Blood Group should only allow:
  - A+
  - A−
  - B+
  - B−
  - AB+
  - AB−
  - O+
  - O−

Design the table.

Q5. A university wants to ensure a student cannot enroll in the same course twice.

Design the table where the table should contain the columns like

1. Student_ID
2. Course_ID
3. Enrollment_Date

Student_ID and Course_ID should be unique and there should not be any NULL values.

Q6. A bank wants to ensure:

- Account Balance can never be negative.
- Account Number is unique.
- Account Type can only be Savings or Current.

Create the tablewith above constraints and the three columns - Account Balance, Account Number, Account Type.

Q7. Create an Enrollment table with the following columns:

Column Name	Data Type	Constraint
EnrollmentID	INT	Primary Key
StudentID	INT	NOT NULL
StudentName	VARCHAR(50)	NOT NULL
CourseID	INT	NOT NULL
CourseName	VARCHAR(50)	NOT NULL
EnrollmentDate	DATE	NOT NULL
EnrollmentStatus	VARCHAR(20)	Default 'In Progress'

Insert the following records into the table:

Enrollment ID	Student ID	stu_name	Course ID	Course Name	Enrollment Date	Enrollment Status
101	1001	Aarav Sharma	201	Python Programming	2026-01-10	In Progress
102	1002	Priya Singh	202	SQL Fundamentals	2026-01-12	In Progress
103	1003	Rahul Verma	203	Power BI	2026-01-15	Completed
104	1004	Sneha Patel	201	Python Programming	2026-01-18	In Progress
105	1005	Arjun Mehta	204	Tableau	2026-01-20	Dropped
106	1006	Neha Gupta	202	SQL Fundamentals	2026-01-22	In Progress

Task

A student has successfully completed the SQL Fundamentals course.

Write an SQL query to update the **EnrollmentStatus** from '**In Progress**' to '**Completed**' for Student **ID 1002.**

Q8. A retail store maintains information about the products it sells.

Create a Products table with the following columns:

Column Name	Data Type	Constraint
ProductID	INT	Primary Key
ProductName	VARCHAR(100)	NOT NULL
Category	VARCHAR(50)	NOT NULL
Brand	VARCHAR(50)	NOT NULL
Price	DECIMAL(10,2)	NOT NULL
Stock	INT	DEFAULT 0

Insert the following records into the table:

Product ID	Product Name	Category	Brand	Price	Stock
101	Laptop	Electronics	Dell	65000.00	15
102	Smartphone	Electronics	Samsung	45000.00	30
103	Office Chair	Furniture	IKEA	8500.00	20
104	Headphones	Electronics	Sony	5000.00	50
105	Study Table	Furniture	IKEA	12000.00	10
106	Smartwatch	Electronics	Apple	35000.00	18
107	Microwave Oven	Electronics	LG	18000.00	12
108	Bookshelf	Furniture	Home Centre	9500.00	8

Task

The store has announced a 20% discount on all products belonging to the Electronics category.

Write an SQL query to reduce the price of all products in the **Electronics** category by 20%.

Q9. An e-commerce company maintains order details in an Orders table.

Create an Orders table with the following data:

Order ID	Customer Name	Product Name	Order Date	Quantity	Total Amount
101	Aarav Sharma	Laptop	2026-07-01	1	65000.00
102	Priya Singh	Smartphone	2026-07-02	2	90000.00
103	Rahul Verma	Headphones	2026-07-03	3	15000.00
104	Sneha Patel	Smartwatch	2026-07-04	1	35000.00
105	Arjun Mehta	Microwave Oven	2026-07-05	1	18000.00

Task

The company wants to track the delivery status of every order.

Write an SQL statement to modify the **Orders** table by adding a new column named **Delivery_Status**. The column should have a default value of '**Pending**' for all new records.

Q10. A hospital wants to maintain records of patient appointments.

Design an **Appointments** table with the following columns:

Column Name	Data Type
AppointmentID	INT
PatientID	INT
DoctorID	INT
AppointmentDate	DATE
AppointmentTime	TIME
Reason	VARCHAR(100)

Task

Create the **Appointments** table by applying the following constraints:

1. **AppointmentID** should be the Primary Key.

2. A doctor cannot have two appointments scheduled at the same date and time.

3. AppointmentID should be the Primary Key.

Q11. A company wants to maintain records of employees working on different projects.

Design an **Employee_Project** table with the following columns:

Column Name	Data Type
AssignmentID	INT
EmployeeID	INT
ProjectID	INT
AssignedDate	DATE
Role	VARCHAR(50)

Task

Create the **Employee_Project** table by applying the following constraints:

1. **AssignmentID** should be the Primary Key.

2. An employee can work on multiple projects.

3. A project can have multiple employees.

4. The same employee should not be assigned to the same project more than once.

4. Choose appropriate data types and constraints for all columns.

Q12. Design a Loans table where:

- Loan amount must be between ₹50,000 and ₹50,00,000.
- Interest rate must be between 5% and 20%.
- Status defaults to 'Pending'.

Q13. Create a Trips table where:

- Trip ID is unique.
- Fare cannot be negative.
- Trip Status can only be:
    - Booked
    - Started
    - Completed
    - Cancelled

Q14. Design a complete Employees table with:

- Employee ID → Primary Key
- Email → Unique
- Salary → Between ₹25,000 and ₹5,00,000
- Department → HR, IT, Finance, Sales only
- Joining Date → Default Current Date
- Employee Name → Mandatory

Q15. A company maintains information about the products it sells. The management has discovered that some product codes are duplicated and wants to prevent this from happening in the future.

The following **Products** table already exists:

Column Name	Data Type
ProductID	INT
ProductCode	VARCHAR(20)
ProductName	VARCHAR(100)
Category	VARCHAR(50)
Price	DECIMAL(10,2)

Task

Write an SQL statement to modify the existing Products table so that every **ProductCode** is unique in future records.